ECE 310 Fall 2026
Lecture 23: Spectral Analysis Part 2

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Learning Objectives

After this lecture, you should be able to:

- Identify the key challenges in performing spectral analysis including the varying amplitudes and distances between sinusoidal components.
- Apply full-lobe and half-lobe resolution criteria to determine the necessary length of signals to successfully perform spectral analysis.
- Describe how the choice of window function can affect spectral analysis.
- Explain the trade-offs between common window functions, e.g. rectangular and Hamming windows.



Spectral Analysis Problem

Spectral analysis

Spectral analysis is a common application of digital signal processing where we seek to identify unknown sinusoidal components of a finite-length signal.

$$x[n] = \sum_{m=1}^{M} A_m \cos(\omega_m n),\ 0 \leq n \leq N-1$$

Unknowns include:

- Number of spectral components M.
- Frequencies $\omega_m,\ 0 \leq m \leq M-1$.
- Amplitudes $A_m,\ 0 \leq m \leq M-1$.



Spectrum of Finite-length Sinusoids

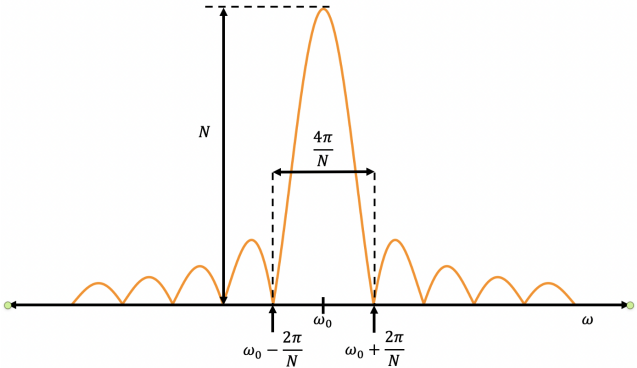
Let $x[n] = A\cos(\omega_0 n),\ 0 \le n \le N-1.$

Then,

$$X_d(\omega) = \frac{A}{2}C(\omega-\omega_0) + \frac{A}{2}C(\omega+\omega_0),$$

where

$$C(\omega) = e^{-j\frac{\omega}{2}(N-1)}\left(\frac{\sin\left(\frac{N}{2}\omega\right)}{\sin\left(\frac{1}{2}\omega\right)}\right)$$



Magnitude spectrum of $C(\omega-\omega_0)$.

Key Spectral Analysis Challenges

Let $s[n] = A_1 \cos(\omega_1 n) + A_2 \cos(\omega_2 n),\ 0 \leq n \leq N-1$.

1. **The frequencies of components may be too close to separate.**
   If $|\omega_1 - \omega_2|$ is too small, we may not be able to distinguish the main lobes from two components as they will combine and appear to form one main lobe.

2. **The amplitudes of components may vary greatly.**
   If amplitude A_2 is much smaller than A_1, e.g. $\frac{1}{10}$ vs. 1. the main lobe of the second component may be covered up by a side lobe of the first component.



Separating Main Lobes

Let $s[n] = 2\cos(\omega_1 n) + 2\cos(\omega_2 n),\ 0 \le n \le N-1$. For $0 \le \omega \le \pi$,

$$
\begin{aligned}
S_d(\omega) &= C(\omega - \omega_1) + C(\omega - \omega_2) \\
|S_d(\omega)| &= \left| \sqrt{C^2(\omega - \omega_1) + C^2(\omega - \omega_2) + 2C(\omega - \omega_1)C(\omega - \omega_2)} \right| \\
&\approx |C(\omega - \omega_1)| + |C(\omega - \omega_2)|
\end{aligned}
$$



Main Lobe Separation Criteria

Lobe separation criteria

For a given pair of spectral components with frequencies ω_1 and ω_2 in a length–N signal, we have two main lobe separation criteria.

- Full lobe separation:

$$N > \frac{4\pi}{|\omega_1 - \omega_2|}$$

- Half lobe separation:

$$N > \frac{2\pi}{|\omega_1 - \omega_2|}$$



Example 1: Lobe Separation

A guitar in standard tuning has musical notes ranging from ≈ 82 Hz to 880 Hz. The two closest notes are at $F_1 = 82$ Hz and $F_2 = 87$ Hz. Suppose you record guitar audio using your phone at a sampling rate of $f_s = \frac{1}{T} = 8$ kHz.

(a) How many samples do you need to resolve any two frequencies (musical notes) in the audio with full lobe resolution? How much time is this?

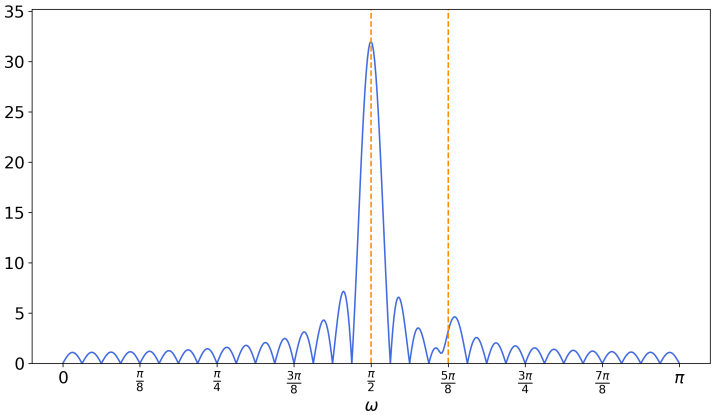
(b) Does the amount of time change if the sampling rate doubles or halves?



Side Lobe Reduction

Consider a case where two components have very different magnitudes.

$$s[n] = \cos(0.5\pi n) + \frac{1}{10}\cos(0.625\pi n),\ 0 \le n \le 63$$



Magnitude spectrum $|S(\omega)|$. Orange dashed lines identify ω_1 and ω_2.

Side Lobe Reduction

Recall how we previously described the DTFT of finite-length signals as the product of an infinite-length signal with a *window function*.



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$$s[n] = \underbrace{(\cos(0.5\pi n) + \frac{1}{10}\cos(0.625\pi n))}_{x[n]} \underbrace{(u[n] - u[n-64])}_{w[n]}$$

$$= x[n]w[n]$$



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$$= x[n]w[n]$$

Thus,

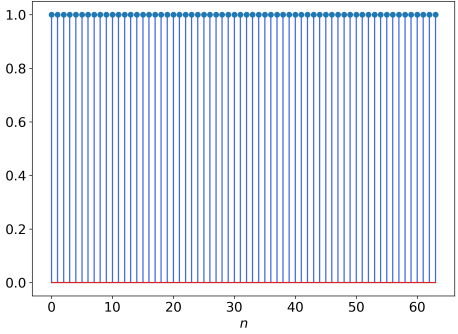
$$s[n] = x[n]w[n] \overset{\mathcal{F}}{\longleftrightarrow} S_d(\omega) = \frac{1}{2\pi}X_d(\omega) * W_d(\omega)$$

$S_d(\omega)$ is then

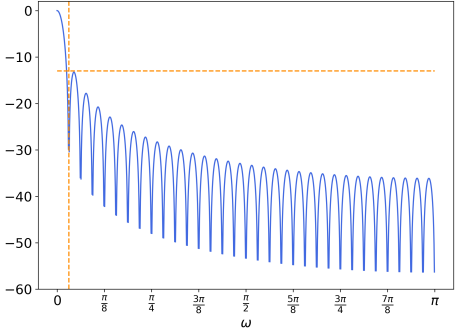
Window Functions

There are many possible choices of window functions and we will explore more later. For now, we compare the *rectangular* and *Hamming* windows.

$$r[n] = \begin{cases} 1, & 0 \le n \le N-1 \\ 0, & \text{otherwise} \end{cases}$$



Rectangular window $r[n]$

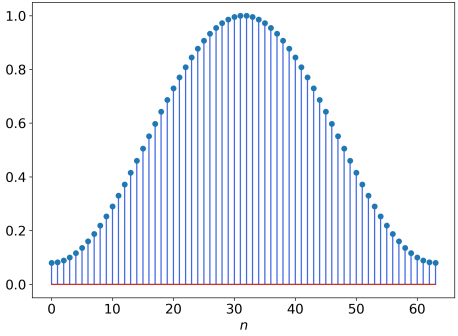


Rectangular window $|R_d(\omega)|$ (log scale)

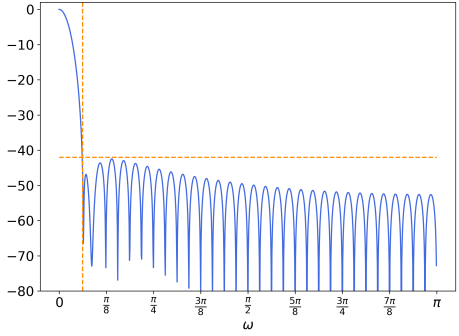
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$$h[n] = \begin{cases} 0.54 - 0.46\cos\left(\frac{2\pi n}{N-1}\right), & 0 \le n \le N-1 \\ 0, & \text{otherwise} \end{cases}$$



Hamming window $r[n]$



Hamming window $|H_d(\omega)|$ (log scale)

Window Function Tradeoff

The rectangular and Hamming windows present a tradeoff between *main lobe width* and *side lobe attenuation/height*. Ideally, we want narrow main lobes, i.e. easier to separate spectral components, and low side lobes, i.e. less likely for side lobes to hide other components.

	Rectangular window	Hamming window
Main Lobe Width	$\frac{4\pi}{N}$	$\frac{8\pi}{N}$
Side Lobe Attenuation	-13 dB	-42 dB

